

DPP -3 : Sexual Reproduction In Flowering Plant

1. Plants developed from apomictic embryos of a plant are

- (1) Genetically different to each other
- (2) Morphologically similar to each other but genetically different from parent plant
- (3) Different to each other w.r.t. their ploidy
- (4) Genetically identical to each other as well as to their parent

2. Which one is correct ?

(A) Pollen tube ruptures inside one of the synergids

(B) Antipodal cell nucleus gets transformed into primary endosperm nucleus (PEN)

- (1) Both A and B are correct
- (2) Both A and B are incorrect
- (3) Only A is correct
- (4) Only B is correct

3. The correct polarity of nucleus and central vacuole in the egg is

- (1) Nucleus towards chalazal end and central vacuole towards micropylar end
- (2) Nucleus towards micropylar end and central vacuole towards chalazal end
- (3) Nucleus and central vacuole both towards micropylar end
- (4) Nucleus and central vacuole both towards chalazal end

4. Which of these is correct?

- (1) Syngamy = Triple fusion + Double fertilization
- (2) Double fertilization = Syngamy + Triple fusion
- (3) Triple fusion = Double fertilization – Syngamy
- (4) More than one option is correct

5. Match the following

Column-I	Column-II
(a) Date palm	(i) Oily
(b) Rice	(ii) Cellulosic
(c) Castor	(iii) Proteinaceous
(d) Ivory palm	(iv) Starchy
(e) Maize	(v) Hemi-cellulosic
(1) a(iv), b(v), c(i), d(ii), e(iii)	
(2) a(v), b(i), c(iv), d(ii), e(iii)	
(3) a(v), b(iv), c(i), d(ii), e(iii)	
(4) a(v), b(iv), c(i), d(iii), e(ii)	

6. The plants in which some of the nucellar cells surrounding the embryo sac start dividing protrude into the embryo sac and develops into the embryos are :-

- (1) Citrus and mango
- (2) Papaya and Citrus
- (3) Citrus and litchi
- (4) Mango and guava

7. The figure shows stages in-



- (1) Embryo development in dicot
- (2) Embryo development in monocot
- (3) Embryo development in gymnosperm
- (4) Both A and B

8. Which of the given statements are related to outbreeding devices.

- (a) Discourage self pollination.
- (b) Prevent loss in genetic variation.
- (c) Different placement of anther and stigma

- (1) a & b only
- (2) a & c only
- (3) b & c only
- (4) All a, b & c

9. Choose odd one with respect to double fertilization and triple fusion

- (1) First studied by Nawaschin in Liliun and Papaver
- (2) It includes syngamy and triple fusion
- (3) Total five nuclei are involved
- (4) It is key character of angiosperms

10. Which one is mismatched?

- (1) True fruit – Mango
- (2) False fruit – Cashewnut
- (3) Parthenocarpic fruit – Banana
- (4) Fleshy fruit – Mustard

11. Archegonium of ovule is :

- (1) Single celled derived from integuments
- (2) Single celled derived from nucellar hypodermis
- (3) Multicellular derived from nucellar epidermis
- (4) Multicellular derived from nucellar hypodermis

12. The given plant shows all adaptations for pollination, except



- (1) Light and non-sticky pollen
- (2) Well exposed stamens
- (3) Presence of nectaries
- (4) Flowers packed into inflorescence

13. Select functions associated with tapetum

- (i) Nutrition to the developing microspores
- (ii) Pollen kit synthesis
- (iii) Sporopollenin and compatibility protein secretion

- (1) (i) & (ii) only
- (2) (ii) only

- (3) (i) & (iii) only
- (4) All (i), (ii) & (iii)

14. Pollination between flowers of different plants of the same species is

- (1) Xenogamy
- (2) Geitonogamy
- (3) Autogamy
- (4) Cleistogamy

15. Anemophilous flowers do not require

- (1) Light weight pollen grains
- (2) Well-exposed stamens
- (3) Feathery stigma
- (4) Nectaries

16. Read the following statement and choose the set of correct statements :-

- (a) Majority of insect pollinated flowers are large.
- (b) Pollination does not guarantee the transfer of right type of pollen.
- (c) Pollination by wind is quite rare in flowering plants and is limited to about 30 genera mostly monocotyledons.
- (d) Both wind and water pollinated flowers are very colourful and produce nectar.
- (e) Opposite the micropylar end, is the chalaza, representing the apical part of the ovule.

Choose the correct answer

- (1) (a), (b) and (c) only
- (2) (a), (b) and (e) only
- (3) (a) and (b) only
- (4) (c), (d) and (e) only

17. The pollens of which of the following plants have the viability period about 30 minutes?

- (1) Rice
- (2) Pea
- (3) Bean
- (4) Brinjal

18. Which of the following does not ensure autogamy?

- (1) Cleistogamy
- (2) Homogamy
- (3) Bud pollination
- (4) Dichogamy

19. Choose the odd one with respect to plants causing pollen allergy

- (1) Chenopodium
- (2) Amaranthus
- (3) Viola
- (4) Parthenium

20. Ploidy level of polar nucleus and definitive nucleus is respectively

- (1) N & 3N
- (2) N & N
- (3) N & 2N
- (4) 2N & 2N

21. Which of the following conditions show that the embryo of a seed is entering the state of dormancy?

- A. Dry weight of seed is increasing.
- B. Water content is reducing.
- C. The general metabolic activity of embryo is slowing down.
- D. Oxygen consumption of seed is increasing.

Choose the correct answer

- (1) A and B only
- (2) A, B and D
- (3) B and D only
- (4) B and C

22. Given below are two statements :

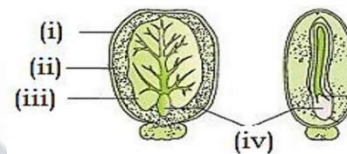
Statement-I : Most of zygotes divide only after certain amount of endosperm is formed.

Statement-II : In the pea family the cotyledon is called scutellum that situated towards one side (lateral) of the embryonal axis.

Choose the most appropriate answer

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II are correct
- (4) Both Statement-I and statement-II are correct

23. Identify the endosperm in the given figure



- (1) I
- (2) II
- (3) III
- (4) IV

24. Match Column-I with Column - II and select the correct option

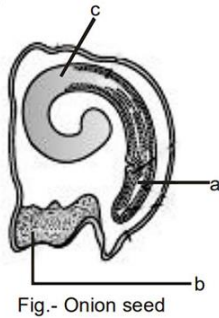
Column - I	Column - II
a) Coleorrhiza	(i) Fruit wall
b) Scutellum	(ii) Covering of radicle
c) Pericarp	(iii) Persistent nucellus
d) Perisperm	(iv) Single cotyledon

- (1) a(ii), b(iv), c(iii), d(i)
- (2) a(iv), b(ii), c(i), d(iii)
- (3) a(ii), b(iv), c(i), d(iii)
- (4) a(iii), b(iv), c(i), d(ii)

25. After double fertilization, most zygotes divide only after certain amount of endosperm is formed. One importance of this adaptation is

- (1) To push the developing embryo out of the micropyle
- (2) To prevent the developing zygote form being divided into many embryos
- (3) To provide assured nutrition to the developing embryo
- (4) To provide sufficient space for developing embryo by dissolving outer and inner integuments of ovule

26. Label the points a, b and c of the figure given below and select the option accordingly



- (1) a-Plumule, b-Endosperm, c-Cotyledon
(2) a-Hypocotyl, b-Seed coat, c-Plumule
(3) a-Hypocotyl, b-Endosperm, c-Cotyledon
(4) a-Coleorrhiza, b-Plumule, c-Cotyledon

27. Find out the correct developmental sequence in the life cycle of flowering plants

- (1) Pollen mother cell megaspore tetrad
(2) Nucellus Megaspore mother cell
(3) Secondary nucleus PEN
(4) Both (2) and (3)

28. Given below are two statements :

- (A) Polar nuclei are situated below the egg apparatus in the large central cell.
(B) Filiform apparatus plays important role in guiding the entry of pollen tube into embryo sac.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (B) are correct
(2) Both (A) and (B) are incorrect
(3) Only (A) is correct
(4) Only (B) is correct

29. Which of the following plants have stony endosperm?

- (1) Castor and groundnut
(2) Areca nut and Phoenix dactylifera
(3) Wheat and coconut
(4) Barley and date palm

30. One of the examples of seeds in which the endosperm is completely consumed during embryo development is

- (1) Pea
(2) Maize
(3) Wheat
(4) Sunflower

31. Arrange microsporangial wall in sequence of outside to inside

- (1) Epidermis, middle layer, endothecium, tapetum
(2) Epidermis, endothecium, middle layer, tapetum
(3) Epidermis, middle layer, tapetum, endothecium
(4) Endothecium, middle layer, tapetum, epidermis

32. Read the following statements and choose the set of correct statements :-

- (a) Pollen grains represents the male gametophytic generation.
(b) Some angiosperms produce more than one embryo in their seed this phenomenon is called polyembryony.
(c) In angiosperm the fruit is the only final product of sexual reproduction.
(d) The portion of embryonal axis above the level of cotyledons is hypocotyl which terminates with the plumule or stem tip.
(e) Embryo development precedes endosperm development.

Choose the correct answer from the options

- (1) (a) and (b) only
(2) (a), (b) and (e) only
(3) (a), (b) and (d) only
(4) (c), (d) and (e) only

33. Read the statements carefully and find how many is/are correct

- (a) Pre-fertilisation involves gametogenesis
(b) Gamete mother cells may be haploid or diploid
(c) Gametogenesis involves mitosis or meiosis
(d) Water is always needed for gamete transfer

- (1) 4
- (2) 3
- (3) 2
- (4) 1

34. Consider the following statements regarding angiosperms and answer accordingly

- (a) Microspore represent the mature male gametophyte.**
- (b) Pollen tablets are used as food supplement**
- (c) Polar nuclei are situated below the egg apparatus.**

Choose the correct answer from the options given below.

- (1) a and b are true but c is false
- (2) b and c are true but a is false
- (3) a and c are true but b is false
- (4) a, b and c are false

35. Given diagram is of –



- (1) Multicarpellary apocarpous gynoecium of michelia
- (2) Multicarpellary syncarpous gynoecium of michelia
- (3) Multicarpellary syncarpous gynoecium of papaver
- (4) Multicarpellary apocarpous gynoecium of papaver

36. In an embryo sac, the largest cell is

- (1) Synergids
- (2) Antipodal cells
- (3) Central cell
- (4) Oosphere

37. Which of the following is wrong ?

- (1) Pollen grains are about 25 μm in diameter
- (2) Exine is the outer layer of pollen
- (3) Intine is made of sporopollenin
- (4) Generative cell is spindle shaped

38. How many given structures are haploid

Ovary, anther, egg, pollen grain, male gamete, zygote

- (1) 3
- (2) 4
- (3) 5
- (4) 6

39. Given below are two statements :

Assertion : Pollen grains can withstand high temperatures and strong acid and alkali.

Reason : Their hard outer layer exine is made up of sporopollenin.

Choose the correct answer from the options given.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (2) (A) is correct but (R) is not correct.
- (3) (A) is not correct but (R) is correct.
- (4) Both (A) and (R) are correct but (R) is the correct explanation of (A).

40. Given below are two statements :

Statement-I: The hard inner layer of pollen grain intine, has prominent apertures called germ pores.

Statement-II: Cytoplasm of pollen grain is surrounded by Tapetum.

Choose the correct answer from the options given.

- (1) Statement-I & II are correct
- (2) Statement I is correct & Statement-II is Incorrect
- (3) Both Statements are Incorrect
- (4) Statement II is only correct

41. What is incorrect :

- (a) In a transverse section, a typical microsporangium appears near circular in outline
- (b) Four wall layers of microsporangium help in dehiscence of anther
- (c) Nucleus of vegetative cell in pollen grain is irregular in shape
- (d) Pollen grains are about 25-50 μ m in diameter

- (1) Only A
- (2) Only B, d
- (3) Only A, B, D
- (4) All

42. How many nucleus Participates in syngamy

(Generative fertilization) and triple fusion (vegetative fertilization) respectively ?

- (1) 3, 2
- (2) 2, 3
- (3) 2, 2
- (4) 3, 3

43. Read the following statements carefully :

- (A) Cleistogamous flowers are invariably autogamous.
- (B) Plant use two abiotic and one biotic agents to achieve pollination.
- (C) Wind pollination is quite common in grasses.
- (D) Pollination by water is quite safe in flowering plant.

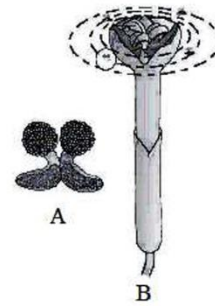
How many of the above statements are correct

- (1) One
- (2) Two
- (3) Three
- (4) Four

44. Monoecious plants prevents :

- (1) Both autogamy and geitonogamy
- (2) Autogamy, but not geitonogamy
- (3) Geitonogamy, but not autogamy
- (4) Neither autogamy nor geitonogamy

45. Following diagram A and B represents ?



- (1) (A) Female flower at Hydrilla (B) Male flower of Hydrilla
- (2) (A) Male flower of Hydrilla (B) Female flower of Hydrilla
- (3) (A) Female flower of Vallisneria (B) Male flower of Vallisneria
- (4) (A) Male flower of Vallisneria (B) Female flower of Vallisneria

46. What does the filiform apparatus do at the entrance of pollen tube into the ovule?

- (1) It brings about opening of the pollen tube
- (2) It guides the pollen tube from a synergid to the egg.
- (3) It helps in the entry of pollen tube into a synergid
- (4) It prevents entry of more than one pollen tube into the embryo sac

47. Given below are two statements :

Assertion :- Night flowering plants are entomophilous.

Reason:- Flowers of these plants are odour less

Choose the correct answer from options given below :

- (1) Both Assertion & Reason are True and the Reason is the correct explanation of the Assertion
- (2) Both Assertion & Reason are True but the Reason is not the correct explanation of the Assertion
- (3) Assertion is True statement but Reason is false
- (4) Both Assertion and Reason are False statements

48. Match the columns :

Column-I

Column-II

(a) Ovary

(i) Fruit

(b) Triple fusion

(ii) Endosperm

(c) Persistent nucellus

(iii) Perisperm

(d) Removal of anther

(iv) Emasculation

(1) a - i, b - ii, c - iii, d - iv

(2) a - ii, b - i, c - iii, d - iv

(3) a - i, b - iii, c - ii, d - iv

(4) a - ii, b - iii, c - i, d - iv

49. Read the following statement and choose correct option :

(i) Angiospermic endosperm is mostly triploid ($3n$)

(ii) Mostly endosperm formation start after first division of zygote

(iii) Coconut has nuclear and cellular endosperm

(iv) Fusion of male gamete and two polar nuclei is called double fertilisation

(1) i, ii and correct and iii, iv incorrect

(2) i, ii, iii correct and iv incorrect

(3) i, iii, correct and ii, iv incorrect

(4) i, ii, iii, iv correct

50. Read the following five statements from (A → E):

A. Majority of flowering plants produce hermaphrodite flowers

B. In sea grasses the female flower reach the surface of water by the long stalk & the male flowers or pollen grains are released on the surface of water.

C. Water pollinated plants Vallisneria & Hydrilla grows in fresh water

D. Wind pollination is rare in grasses

E. Chasmogamous flowers produce assured seed set even in absence of pollinators

How many of above statements are wrong :

(1) Four

(2) Three

(3) Two

(4) Five

Answer Key

1. (4)
2. (3)
3. (1)
4. (4)
5. (3)
6. (1)
7. (1)
8. (4)
9. (1)
10. (4)
11. (2)
12. (3)
13. (4)
14. (1)

15. (4)
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47. (3)
48. (1)
49. (1)
50. (1)